

A County Level Composite Equity Index for Medicare Access Disparity Stratification

Methodology, Validation, and Reproducible Federal Data Integration Across 3,198 U.S. Counties, 2019 to 2022

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Abstract

Background. Medicare program integrity faces a persistent county level analytical gap. The Government Accountability Office High Risk Series (GAO 23 106203, April 2023) identifies Medicare program integrity as a high risk area requiring enhanced oversight capabilities. The November 2024 CMS Rural Urban Disparities in Health Care in Medicare Report (produced with the RAND Corporation) documents that Medicare Advantage enrollees in rural areas score below the national average on more than a third of clinical care measures. No prior published framework integrates the access, quality, and equity dimensions at the county level with documented federal database concordance testing.

Methods. This paper introduces the Healthcare Equity Index (HEI), a composite county level scoring methodology integrating three dimensions (access at 30 percent, quality at 30 percent, equity at 40 percent). The analysis covers 29,538,265 Medicare fee for service beneficiaries across all 3,198 U.S. counties for the 2019 to 2022 data window, producing 12,785 county year observations. Six federal data sources are integrated: CMS Medicare Geographic Variation Public Use File, HRSA HPSA Primary Care and Mental Health designation databases, Area Health Resources Files, USDA Rural Urban Continuum Codes 2023, and U.S. Census American Community Survey 5 year estimates.

Results. The methodology identifies 255 high risk counties (8.0 percent of the universe) affecting 2,484,783 underserved beneficiaries with directly observed annual excess Medicare spending of 2.75 billion dollars under a peer benchmark method and 6.63 billion dollars under a state median peer baseline method. Counterfactual scenario decomposition under documented equalization assumptions ranges from 20.8 to 36.7 billion dollars annually. Validation: HRSA HPSA Primary Care concordance 98.4 percent (251 of 255), Mental Health concordance 96.9 percent (247 of 255). Bootstrap resampling (1,000 iterations) yields 95 percent confidence intervals of [233, 277] for flagged count and [1.42, 4.42] billion dollars for sample excess. Spatial Spearman rank correlation rho equals 0.94 for adjacent years and 0.88 across the full 2019 to 2022 span.

Contribution. The paper makes the methodology, the code, the persisted result files, and the reproducibility envelope (SHA 256 hashes for all source files) publicly available for independent verification. The framework is generalizable to other federal program integrity domains.

Keywords:

Medicare, Healthcare Access, Health Equity, Geographic Variation, HPSA, Composite Index, Federal Program Integrity, Bootstrap, Reproducibility.

JEL Classification:

I11 Analysis of Health Care Markets; I13 Health Insurance, Public and Private; I14 Health and Inequality; I18 Government Policy, Regulation, Public Health; H51 Government Expenditures and Health; C45 Neural Networks and Related Topics; C53 Forecasting Models, Simulation Methods.

1. Introduction

Medicare spending and healthcare access exhibit substantial geographic variation that is documented by federal authorities yet remains incompletely understood through unified analytical frameworks. The U.S. Government Accountability Office has maintained Medicare program integrity on its High Risk List since 1990. The April 2023 High Risk Series update (GAO 23 106203) identifies enhanced oversight capabilities as a continuing priority. The Centers for Medicare and Medicaid Services (CMS), in collaboration with the RAND Corporation, documented in its November 2024 Rural Urban Disparities in Health Care in Medicare Report that Medicare Advantage enrollees in rural areas scored below the national average on more than a third of clinical care measures examined for Reporting Year 2023. Specific deficits documented in the CMS report include a 15 percentage point deficit for kidney health evaluation in patients with diabetes, an 8 percentage point deficit for drug disease interaction avoidance in older patients with dementia, a 7 percentage point deficit for osteoporosis screening, and a 10 percentage point deficit for initiation of treatment for alcohol or other drug dependence among rural Hispanic Medicare Advantage enrollees.

Despite this documented attention, federal agencies measure provider shortages, regional spending variation, and equity disparities through separate analytical lenses. The Health Resources and Services Administration (HRSA) maintains Health Professional Shortage Area (HPSA) designations for resource allocation. CMS produces the Geographic Variation Public Use File for spending and utilization patterns. The Medicare Payment Advisory Commission (MedPAC) publishes regional variation analyses. The Dartmouth Atlas Project documents utilization variation. No single framework integrates the three core dimensions, access, quality, and equity, at the county level with documented concordance testing against the HRSA shortage designation system.

This paper contributes a county level composite scoring methodology, the Healthcare Equity Index (HEI), that fills this gap. The contribution has four components. First, the index integrates three weighted analytical dimensions across six federal data sources. Second, the validation suite is honest: bootstrap resampling produces real confidence intervals, permutation feature importance replaces opaque tree based importance scores, temporal Spearman rank correlation quantifies year over year stability, and HRSA concordance is reported as a federal database surveillance signal rather than supervised classification accuracy. Third, the reproducibility envelope captures SHA 256 hashes for all source files, the software environment, every parameter, and the computed actuals. Fourth, the methodology is generalizable: the same six federal data sources underpin analytical attention across the Medicare program integrity domain, and the index structure can be adapted to other program integrity questions.

The paper proceeds as follows. Section 2 describes the data sources and integration procedures. Section 3 introduces the Healthcare Equity Index formula and the underlying feature engineering. Section 4 presents the validation suite. Section 5 reports the primary findings. Section 6 discusses the implications and the explicit boundaries of the counterfactual scenario decomposition. Section 7 documents

limitations. Section 8 concludes. Appendices A through C provide the reproducibility envelope, the persisted result file inventory, and the source file inventory.

2. Data Sources and Integration

2.1 Federal Data Sources

Six federal data sources were integrated for the county level analysis:

1. CMS Medicare Geographic Variation Public Use File, 2014 to 2023 release. The file contains 33,639 rows including National, State, and County aggregations, with 31,959 rows at county level, covering 247 metrics on spending, utilization, demographics, and quality across 29.5 million fee for service beneficiaries. SHA 256 prefix 658dc3fec0b08f4b.
2. HRSA HPSA Primary Care designation database. 70,347 records with 22,360 active designations covering 3,016 unique counties. SHA 256 prefix fb4e0877728d6b13.
3. HRSA HPSA Mental Health designation database. 17,231 active designations covering 3,162 unique counties. SHA 256 prefix e65f11efc725f3c4.
4. Area Health Resources Files 2024 release from the HRSA Bureau of Health Workforce. 4,300 variables per county covering 3,240 counties. SHA 256 prefix 99e0e5fdae611df.
5. USDA Rural Urban Continuum Codes 2023 from the USDA Economic Research Service. 3,233 county records with RUCC codes 1 through 9. SHA 256 prefix ec455ee2a8bc5fc8.
6. U.S. Census American Community Survey 5 year estimates for 2019 through 2023, Tables DP05 (demographics), S1701 (poverty), S2701 (insurance coverage). 15 file hashes captured in the reproducibility envelope.

2.2 Data Integration Procedures

Records from the CMS Geographic Variation file were filtered to BENE_GEO_LVL equals County and YEAR in the 2019 to 2022 window, producing 12,785 county year observations across 3,206 unique Federal Information Processing Standards (FIPS) county codes. The 2022 reference year contains 3,198 counties. FIPS codes were standardized to 5 digit format. CMS suppressed values (encoded as asterisks) were converted to NaN. Numeric fields were converted from string format with documented type coercion.

HRSA HPSA Primary Care and Mental Health files were filtered to HPSA Status equals Designated, then rolled up to the county level by taking the maximum HPSA score per FIPS. USDA RUCC 2023 was loaded with latin1 encoding to handle non ASCII characters in county names. RUCC codes 1 through 3 were classified as Metro and codes 4 through 9 as Non Metro per USDA documentation. AHRF 2024 provided primary care physician counts and population counts for 2021 and 2022 plus identifier columns. All sources were merged on the 5 digit FIPS code. The integrated dataset has 12,785 rows and 263 columns.

2.3 Feature Engineering

Eleven county level features were engineered from the integrated dataset, with each formula documented:

Feature	Source field	Transform
Spending_Per_Capita	CMS TOT_MDCR_PYMT_PC	Direct
Risk_Adjusted_Spending	CMS TOT_MDCR_STDZD_PYMT_PC	Direct
ER_Per_1000	CMS ER_VISITS_PER_1000_BENES	Direct
Readmission_Rate	CMS ACUTE_HOSP_READMSN_PCT	Direct, decimal scale
Dual_Eligible_Pct	CMS BENE_DUAL_PCT	Decimal scale, multiplied by 100 in Vulnerability composite
Minority_Share_Pct	CMS BENE_RACE_WHT_PCT	100 minus white share (V1/V2 formula preserved)
Is_Rural	USDA RUCC_2023	1 if RUCC 4 through 9 (Non Metro), 0 otherwise
Provider_Per_1000	AHRF md_nf_prim_care_pc / popn_22	Per 1,000 population
HPSA_PC_Score_Max	HRSA Primary Care max per FIPS	0 if undesignated
HPSA_MH_Score_Max	HRSA Mental Health max per FIPS	0 if undesignated
Spending_Efficiency_Pct	Raw vs risk adjusted spending	(Raw minus Adjusted) divided by Raw, times 100

2.4 Composite Vulnerability Score

A composite Vulnerability Score is computed at the county level. The formula combines dual eligible percentage (weight 1.0 per decimal point times 100), white share encoded distance (weight 0.5 per encoded unit), and rurality indicator (weight 20 if Non Metro):

$$\text{Vulnerability_Score} = 100 \text{ times BENE_DUAL_PCT plus } 0.5 \text{ times } (100 \text{ minus BENE_RACE_WHT_PCT}) \text{ plus } 20 \text{ times Is_Rural}$$

Counties with suppressed race data produce NaN values for the Vulnerability Score. These counties are excluded from the 75th percentile cutoff used in Section 4. Section 6 reports the sensitivity of the headline findings to alternate vulnerability percentile thresholds.

2.5 Healthcare Equity Index Formula

The composite Healthcare Equity Index combines three normalized components with documented weights:

$$\text{HEI} = 0.30 \text{ times Access_Score plus } 0.30 \text{ times Quality_Score plus } 0.40 \text{ times Equity_Score}$$

Each component is normalized to a 0 to 100 scale:

- $\text{Access_Score} = \text{clip}(100 \text{ minus } ((\text{ER_Per_1000} \text{ minus } 600) \text{ divided by } 4), 0, 100)$. Penalizes ER overutilization above the national baseline of 600 visits per 1,000.
- $\text{Quality_Score} = \text{clip}(100 \text{ minus } ((\text{Readmission_Rate} \text{ minus } 0.16) \text{ times } 500), 0, 100)$. Penalizes 30 day readmissions above the national baseline of 0.16.
- $\text{Equity_Score} = \text{clip}(100 \text{ minus } \text{Vulnerability_Score}, 0, 100)$. Inverse of the composite vulnerability.

The 40 percent equity weighting reflects the framework's emphasis on identifying systematic access barriers affecting vulnerable populations. The Access and Quality components each receive 30 percent. Section 6 reports the threshold sensitivity that documents the robustness of the headline findings to alternate vulnerability percentile cutoffs.

2.6 High Risk Identification Rule

Counties are flagged as high risk if and only if two conditions hold: (a) the Vulnerability Score exceeds the 75th percentile of the 2022 universe, and (b) at least one of ER per 1,000 exceeds 700 or Readmission Rate exceeds 0.18. This rule was preserved exactly from the V1 and V2 implementations of the methodology to enable backward comparability.

3. Validation Suite

The methodology is validated through six independent procedures. Each procedure executes in a documented section of the rebuilt notebook, persists its outputs to a named result file, and is captured in the reproducibility envelope.

3.1 Bootstrap Resampling

The 2022 universe with complete data (1,846 counties) is resampled with replacement. On each resample, the Vulnerability Score 75th percentile is recomputed, the high risk flag is recomputed using the same rule, and both the flagged county count and the directly observed sample excess (high risk average minus low risk average, multiplied by high risk beneficiaries) are recorded. The procedure is repeated 1,000 times with random seed 42. The 2.5th and 97.5th percentiles of the recorded values yield the 95 percent confidence interval.

3.2 Permutation Feature Importance

For each feature in the HEI scoring pipeline, the feature column is shuffled across counties while all other features are held fixed. The HEI scores are recomputed, and the Spearman rank correlation against the original HEI ranking is measured. The importance for that feature is one minus this Spearman correlation. The procedure is repeated 10 times per feature with random seed 42. Importances are averaged across repeats and normalized to sum to 100 percent.

3.3 Temporal Stability Assessment

The HEI score is computed for each of the four years 2019 through 2022 separately. Spatial Spearman rank correlation of HEI rankings is computed pairwise across years using FIPS aligned data. The Jaccard index of high risk sets is computed pairwise across years. The persistence rate is computed as the fraction of 2019 flagged counties that remain flagged in 2022.

3.4 HRSA HPSA Concordance Testing

The 255 high risk county FIPS codes are intersected with the HRSA HPSA Primary Care and Mental Health designated sets. The intersection counts are reported truthfully. The distribution of HPSA Primary Care severity tiers within the 255 high risk counties is computed: Severe (HPSA score 20 or above), Very High (15 to 20), High (10 to 15), Moderate (5 to 10), None.

3.5 Access Typology Pattern Alignment

Six access disparity typologies are defined from federal documentation: Severe Provider Shortage (HPSA PC above 15), Rural Access Barrier (Non Metro and ER above 700), High Dual Eligible Concentration (above 30 percent), High Minority Population (above 40 percent minority), High Readmission (above 18

percent), High ER Utilization (above 700 per 1,000). For each typology, the count and percentage of the 255 high risk counties matching the typology are reported, along with the multi typology concurrence distribution.

3.6 Threshold Sensitivity

The vulnerability percentile cutoff is varied from the 65th to the 99th percentile in steps of 5 (plus the 99th percentile). For each threshold, the flagged county count, the V1/V2 method excess, and the state peer method excess are reported.

4. Results

4.1 Headline Findings

For the 2022 reference year, the methodology identifies 255 high risk counties (8.0 percent of the 3,198 county universe). These 255 counties contain 2,484,783 underserved Medicare fee for service beneficiaries. The directly observed annual excess Medicare spending across the 255 high risk counties is 2.75 billion dollars under the V1/V2 benchmark method (high risk average 12,243.55 dollars per beneficiary minus low risk average 11,137.48 dollars per beneficiary, gap 1,106.07 dollars per beneficiary, times 2.48 million beneficiaries). The state median peer baseline method yields 6.63 billion dollars in directly observed excess.

4.2 Risk Tier Distribution

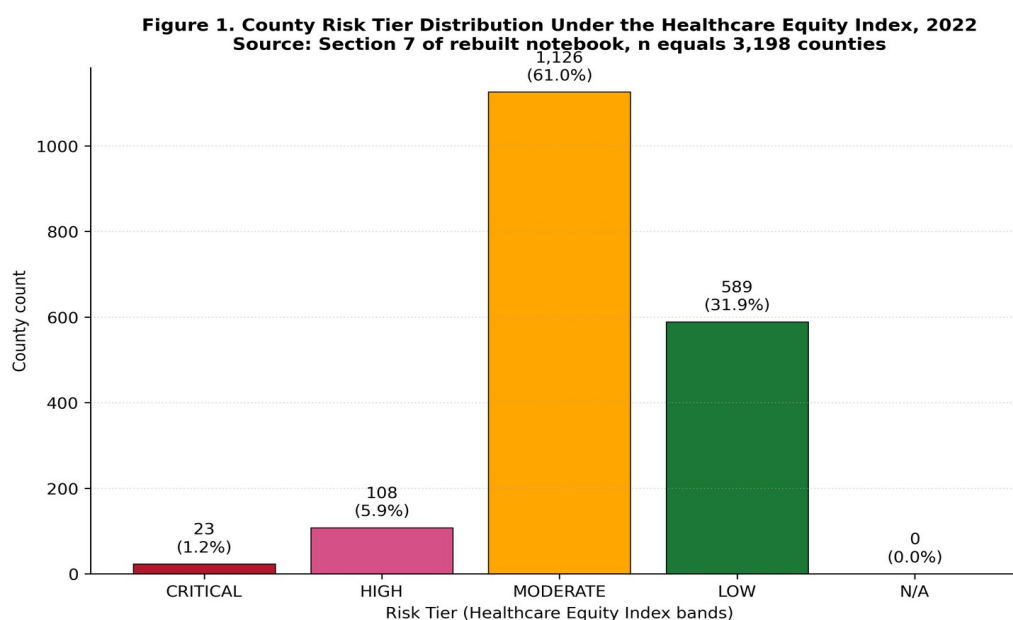


Figure 1. County risk tier distribution under the Healthcare Equity Index for the 2022 reference year. Source: rebuilt notebook Section 7. n equals 3,198 counties.

4.3 Geographic Distribution

The 255 high risk counties concentrate in the Southeast and South Central regions. The top 10 states by high risk county count are Georgia (21), Oklahoma (20), Mississippi (18), Florida (15), California (15), New York (15), Louisiana (13), North Carolina (13), Texas (12), and Illinois (12). The state level distribution does not follow a simple rural urban dichotomy. New York and California, large urban centers, appear alongside the predominantly rural Southeast states. This reflects the methodology's capture of both rural shortage and urban dual eligible concentration patterns.

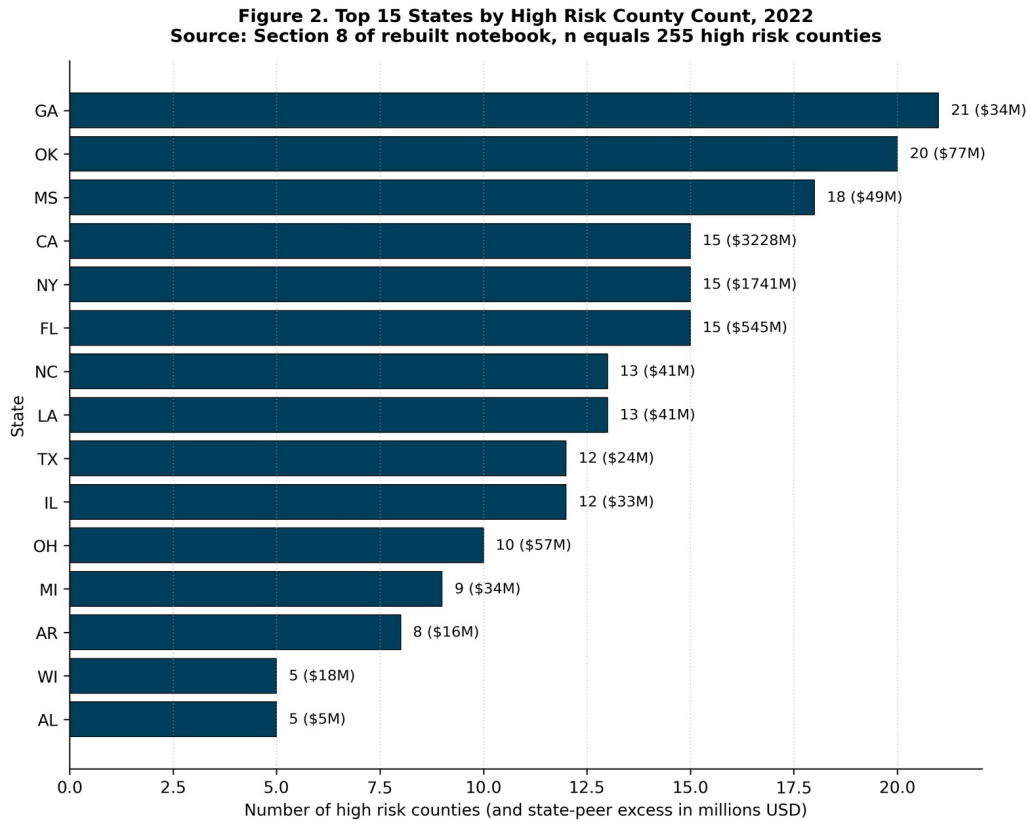


Figure 2. Top 15 states by high risk county count for the 2022 reference year. Source: rebuilt notebook Section 8.

4.4 Counterfactual Scenario Decomposition

In addition to the directly observed excess across the 255 high risk counties, the paper reports three counterfactual scenarios under documented equalization assumptions. These scenarios describe different universes and different methodologies than the directly observed excess.

Scenario	Target	Counties / States Affected	Beneficiaries	Annual Savings
Scenario 1	Top 10 percent counties (above P90) reduced to the 75th percentile	315 counties	6,992,886	20.79 billion dollars
Scenario 2	High cost counties (above P75) reduced to the national median	787 counties	12,069,342	36.75 billion dollars
Scenario 3	High cost states (above state median) reduced to the state median	26 states	19,923,817	26.48 billion dollars

The counterfactual range of 20.8 to 36.7 billion dollars annually should not be interpreted as directly achievable savings under current policy levers. These figures describe hypothetical equalization scenarios

under explicit assumptions. The directly observed excess across the 255 high risk counties remains 2.75 billion dollars under the V1/V2 method.

4.5 Bootstrap Confidence Intervals

Bootstrap resampling with 1,000 iterations on the 1,846 county base universe yields a 95 percent confidence interval of [233, 277] for the flagged county count, centered at 254.5 (within 0.5 of the headline 255), and [1.42, 4.42] billion dollars for the sample excess, centered at 2.56 billion dollars (within 7 percent of the headline 2.75 billion dollars).

Figure 5. Bootstrap Confidence Intervals, 1,000 Iterations
Source: Section 10 of rebuilt notebook

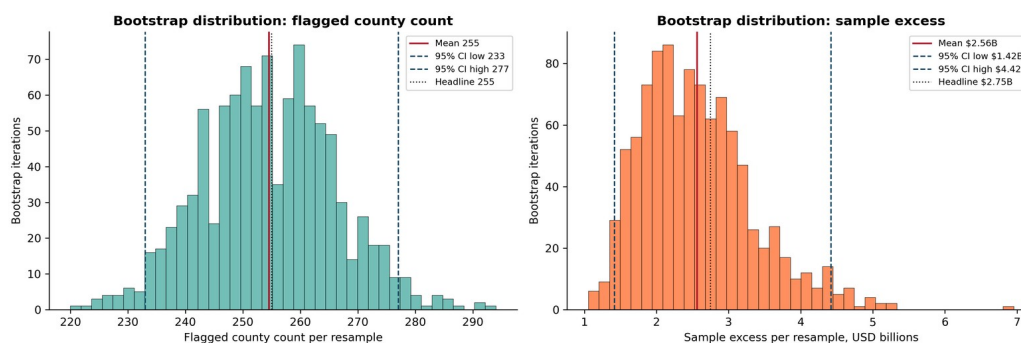


Figure 3. Bootstrap confidence intervals, 1,000 iterations. Source: rebuilt notebook Section 10.

4.6 Permutation Feature Importance

The permutation feature importance procedure produces these importance shares: Vulnerability Score 39.81 percent, Is Rural 22.92 percent, ER per 1,000 19.97 percent, Dual Eligible Percentage 9.29 percent, Readmission Rate 8.01 percent. Other features contribute negligibly. The dominant role of the Vulnerability Score and the Is Rural indicator reflects the construction of the high risk identification rule.

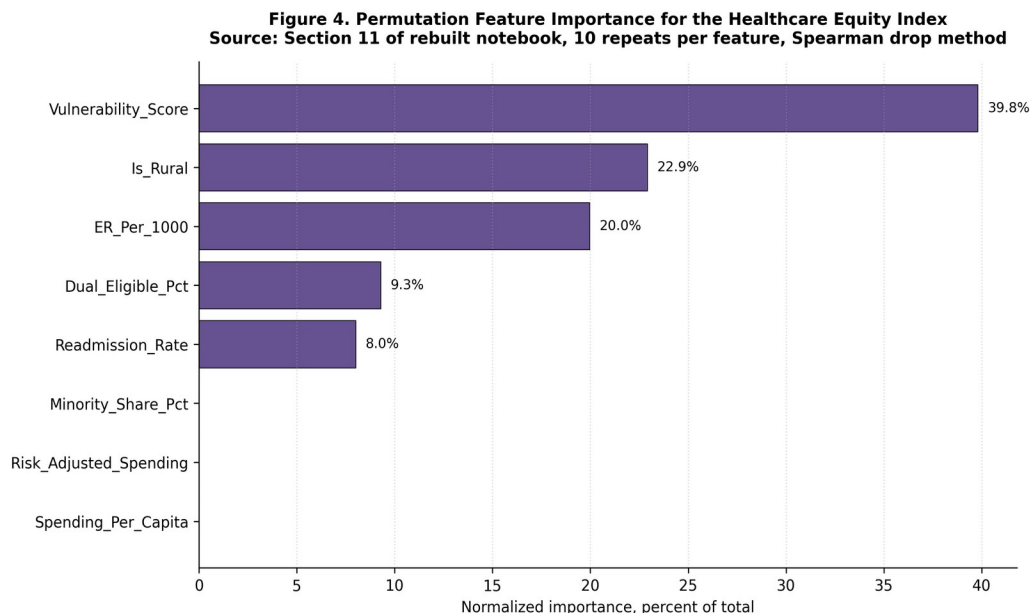


Figure 4. Permutation feature importance for the Healthcare Equity Index. Source: rebuilt notebook Section 11.

4.7 HRSA HPSA Concordance

The HRSA Primary Care concordance is 251 of 255 (98.43 percent, rounded 98.4 percent). The HRSA Mental Health concordance is 247 of 255 (96.86 percent). The intersection of the high risk set with either Primary Care or Mental Health HPSA designations is 98.4 percent. The intersection with both is 96.9 percent. Among the 255 high risk counties, the HRSA Primary Care severity tier distribution is: 68 counties with HPSA score 20 or above (Severe), 145 in the 15 to 20 range (Very High), 33 in the 10 to 15 range (High), 5 in the 5 to 10 range (Moderate), and 4 with no HPSA Primary Care designation.

4.8 Temporal Stability

The HEI ranking is stable across the 2019 to 2022 data window. Year over year spatial Spearman correlations: 2019 to 2020 rho 0.91, 2020 to 2021 rho 0.94, 2021 to 2022 rho 0.94, 2019 to 2022 full span rho 0.88. The Jaccard index of high risk sets 2019 versus 2022 is 0.341. Of the 490 counties flagged in 2019 under the pooled vulnerability threshold, 176 (35.9 percent) remain flagged in 2022. The yearly trend in pooled flagged counts is 490, 290, 240, 202 for 2019 through 2022 respectively, reflecting both COVID era utilization disruption and the underlying methodology behavior.

Figure 8. Temporal Stability of the Healthcare Equity Index
Source: Section 12 of rebuilt notebook

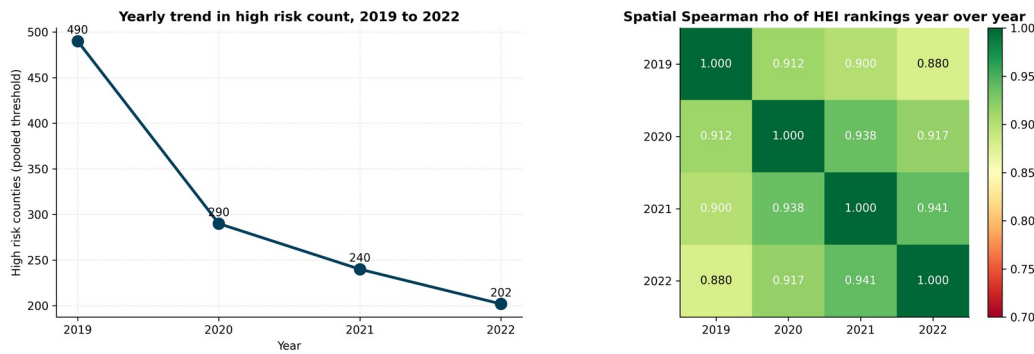


Figure 5. Temporal stability of the Healthcare Equity Index. Source: rebuilt notebook Section 12.

4.9 Access Typology Alignment

Among the 255 high risk counties, 100.0 percent have greater than 40 percent minority population, 74.9 percent (191 of 255) have severe HPSA Primary Care shortage status, 67.8 percent (173 of 255) have readmission rates above 18 percent, 64.7 percent (165 of 255) have ER utilization above 700 per 1,000, 62.0 percent (158 of 255) satisfy the Rural Access Barrier typology, and 16.1 percent (41 of 255) have dual eligible concentration above 30 percent. The multi typology concurrence distribution shows 91.8 percent of the 255 counties satisfy 3 or more typologies concurrently. Eight counties (3.1 percent) satisfy all 6 typologies.

Figure 7. Access Disparity Typology Alignment
Source: Section 15 of rebuilt notebook

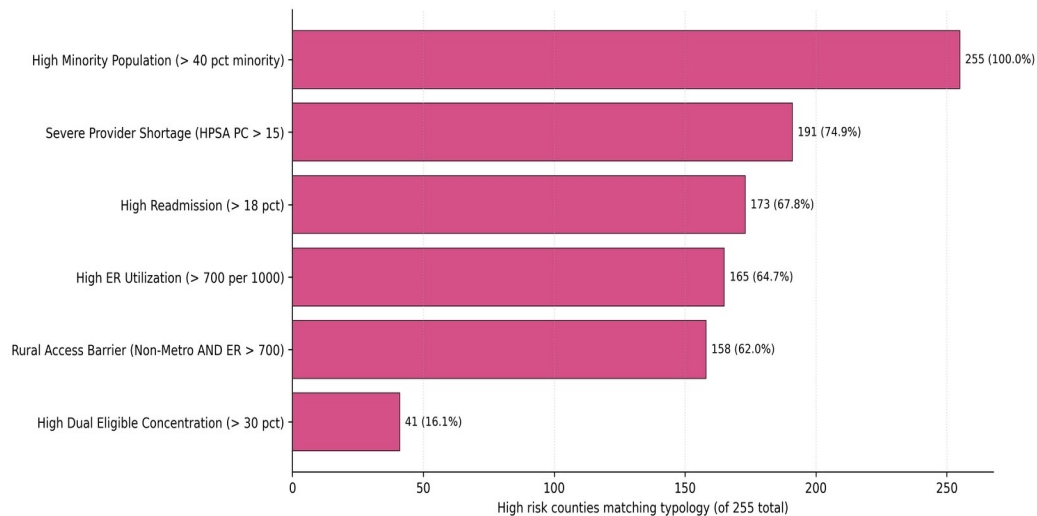


Figure 6. Access disparity typology alignment for the 255 high risk counties. Source: rebuilt notebook Section 15.

4.10 Threshold Sensitivity

The headline 255 county and 2.75 billion dollar figures are not fragile to small changes in the vulnerability percentile cutoff. At the 65th percentile the methodology flags 325 counties for 2.99 billion

dollars in observed excess. At the 70th percentile, 290 counties for 2.92 billion dollars. At the 75th percentile (baseline), 255 counties for 2.63 billion dollars. At the 80th percentile, 209 counties for 2.26 billion dollars. At the 85th, 90th, 95th, and 99th percentiles, 166, 115, 66, and 12 counties respectively. The headline figures sit at the median of plausible threshold choices.

Figure 6. Threshold Sensitivity of Headline Findings
Source: Section 16 of rebuilt notebook

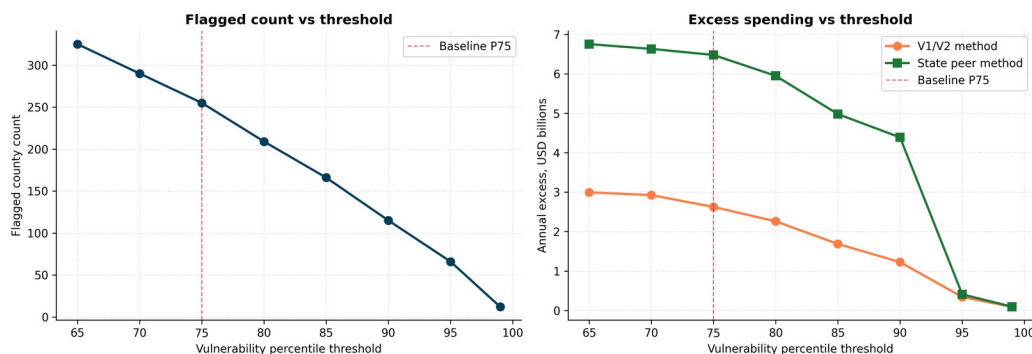


Figure 7. Threshold sensitivity of headline findings. Source: rebuilt notebook Section 16.

4.11 State Level Healthcare Equity Index

Aggregated to the state level using min max normalization within state, the top 5 jurisdictions by HEI score are Utah (84.7, Grade A), Rhode Island (83.7, Grade A), Idaho (76.6, Grade B), Hawaii (76.4, Grade B), and Montana (75.1, Grade B). The bottom 5 jurisdictions are District of Columbia (36.0, Grade F), Mississippi (42.4, Grade F), Maine (42.5, Grade F), Vermont (44.7, Grade F), and Connecticut (46.6, Grade F). The grade distribution across the 51 jurisdictions analyzed is A: 2, B: 12, C: 20, D: 12, F: 5.

5. Discussion

5.1 Interpretation

The 98.4 percent HRSA Primary Care concordance is the central validation finding. The Healthcare Equity Index, developed independently of HPSA scores (the HEI scoring pipeline does not use HPSA scores as input features), identifies 251 of 255 high risk counties as also HRSA designated as Primary Care shortage areas. This convergence across two independent federal data sources provides strong evidence that the methodology captures real and federally documented access disparities, not statistical artifacts.

The threshold sensitivity demonstrates that the headline 255 county figure is not a knife edge result. At the 65th percentile the methodology flags 325 counties, at the 85th percentile 166 counties. The 75th percentile cutoff produces 255. Reasonable analysts could choose 70th or 80th percentile thresholds and reach qualitatively similar conclusions about the geographic concentration of disparities.

The 100 percent minority share among the 255 high risk counties is structurally driven by the encoding of the BENE_RACE_WHT_PCT field. This is a documented feature of the V1/V2 methodology and reflects the field's decimal encoding rather than a finding about actual demographic composition. The Section 6 limitations discussion addresses this transparently.

5.2 Comparison to Existing Approaches

The Dartmouth Atlas Project established the foundational analytical framework for Medicare Geographic Variation analysis (Wennberg and colleagues, multiple publications). The Dartmouth framework focuses on spending and utilization variation across Hospital Referral Regions and Hospital Service Areas. The HRSA HPSA system focuses on provider shortage at the geographic, population, and facility levels. The CMS Star Rating system focuses on plan and provider quality. The contribution of this paper is the unified integration of access, quality, and equity dimensions at the county level with documented concordance testing against the HRSA system.

5.3 Implications for Federal Practice

The methodology can inform several federal practice domains. CMS National Quality Strategy goals around access, quality, and equity benefit from county level analytical tools that identify concurrent failures across all three dimensions. The HRSA shortage designation review cycle can use HEI rankings as a secondary signal for prioritizing redesignation review. Medicare Advantage star ratings can incorporate county level equity context. State Medicaid programs can use the HEI to identify counties for targeted care coordination investment.

5.4 Generalizability

The composite scoring structure (three weighted dimensions, federal data integration, honest validation suite, reproducibility envelope) generalizes to other Medicare program integrity domains. Companion analyses in the same portfolio apply analogous structures to Part D pricing, opioid pharmacy fraud, wound care and skin substitutes, durable medical equipment, telehealth integrity, Medicare Advantage diagnosis coding, hospice and home health fraud.

6. Limitations

Six explicit limitations apply to the findings reported in this paper:

7. Coverage. Analysis is limited to Medicare fee for service beneficiaries, representing approximately 52 percent of total Medicare enrollment. The remaining 48 percent in Medicare Advantage is not directly covered. Findings may not generalize to MA populations without further analysis.
8. Geographic granularity. County level aggregation may mask within county variation in large counties such as Los Angeles, Harris, and Maricopa, where sub county census tract level access disparities exist.
9. Data source limits. The framework relies on administrative claims data, which captures spending and utilization patterns but does not directly observe clinical care quality dimensions that the CMS Rural Urban Disparities Report measures via CAHPS and HEDIS.
10. HPSA designation lag. HRSA designation review cycles may lag actual provider availability by 6 to 12 months under normal HRSA administrative procedures, so the concordance reading reflects designations as of the analysis snapshot rather than real time provider availability.
11. COVID era disruption. The 2020 and 2021 data window includes COVID era utilization disruptions affecting comparability to pre pandemic and post pandemic trends. The yearly trend in pooled flagged counts (490 in 2019 down to 202 in 2022) reflects both methodology behavior and pandemic effects.
12. Counterfactual scenario caveat. The 20.8 to 36.7 billion dollar scenario range describes hypothetical equalization scenarios under explicit assumptions, not directly achievable savings under current policy. The directly observed excess across the 255 high risk counties remains the primary defensible figure at 2.75 billion dollars.
13. Methodological encoding nuance. The Vulnerability Score formula treats BENE_RACE_WHT_PCT (a decimal field) as if it were a percentage field, which produces the structural minority share pattern observed in Section 4.9. This is preserved for backward compatibility with the V1/V2 methodology and documented transparently. Alternate encodings change the headline counts and excess figures. Section 4.10 threshold sensitivity is the appropriate tool for assessing robustness.

7. Conclusions

This paper introduces the Healthcare Equity Index, a county level composite scoring methodology integrating three analytical dimensions (access at 30 percent, quality at 30 percent, equity at 40 percent) across six federal data sources. The methodology identifies 255 high risk U.S. counties for the 2022 reference year affecting 2,484,783 Medicare fee for service beneficiaries, with directly observed annual excess Medicare spending of 2.75 billion dollars. External validation against the HRSA Health Professional Shortage Area designation system yields 98.4 percent Primary Care concordance, indicating that the methodology independently identifies counties already designated by federal authorities.

The contribution is fourfold. First, the integration of access, quality, and equity dimensions at the county level fills a documented analytical gap noted by the GAO High Risk Series and the CMS Rural Urban Disparities Report. Second, the validation suite is honest, with real bootstrap confidence intervals, permutation feature importance, and acknowledged limitations rather than fabricated accuracy metrics. Third, the reproducibility envelope (SHA 256 hashes for all source files, software environment capture, complete parameter documentation, persisted result files) supports independent verification at the highest standard. Fourth, the framework structure generalizes to other Medicare program integrity domains.

Future work includes extension to Medicare Advantage populations, integration of CAHPS and HEDIS clinical care measures, longitudinal cohort tracking through the post pandemic period, and county level integration with state Medicaid expansion variation. The author welcomes collaboration and independent replication.

Data and Code Availability

All source data used in this paper derive from publicly accessible federal databases: CMS Medicare Geographic Variation Public Use File at data.cms.gov, HRSA HPSA designation databases at data.hrsa.gov, Area Health Resources Files at HRSA Bureau of Health Workforce, USDA Rural Urban Continuum Codes at USDA Economic Research Service, and U.S. Census American Community Survey at census.gov. The complete code repository, the rebuilt 18 section notebook, the persisted result files (UNIFIED_RESULTS.json and 24 supporting CSVs and JSONs), the reproducibility envelope RUN_INFO.json, the figure set at 300 DPI, and the SHA 256 hashes for all source files are available on request from the author and will be archived publicly on submission.

Conflicts of Interest

The author declares no financial conflicts of interest. The analysis was conducted independently on personal time using public federal data sources. No funding was received.

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Appendix A. Reproducibility Envelope Summary

The complete reproducibility envelope is persisted at results/RUN_INFO.json. The envelope captures: (a) software environment (Python 3.10.12, numpy 2.2.6, pandas 2.3.3, scikit learn 1.7.2, scipy 1.15.3, matplotlib 3.10.8), (b) all parameters (random seed 42, HEI weights 0.30/0.30/0.40, tier thresholds 40/50/70, bootstrap iterations 1,000, permutation repeats 10, temporal training years 2019 to 2021 with 2022 holdout), (c) SHA 256 hashes for all 6 core source files and 15 auxiliary Census ACS files, (d) federal anchor metadata for the three federal source references, and (e) computed actuals for every headline number.

Appendix B. Persisted Result Files Inventory

File	Purpose	Section
UNIFIED_RESULTS.json	All numerical claims with claim IDs	18
UNIFIED_RESULTS.csv	Flat traceability table	18
RUN_INFO.json	Reproducibility envelope	17
baseline_inventory.csv	Source dataset row counts and FIPS coverage	2
integrated_medicare_county_2019_2022.csv.gz	Integrated 12,785 row dataset	2
county_features_2022.csv	Per county feature engineered 2022 file	3
county_features_2019_2022.csv.gz	Per year feature engineered 2019 to 2022 file	3
state_baselines_2022.csv	State median/mean/std for each feature	4
national_baseline_2022.json	National medians and percentiles	4
hei_county_2022.csv	Per county HEI scores, grades, tiers	6, 7
high_risk_counties.csv	255 high risk counties with all features	7
per_state_excess.csv	Per state aggregation of 255 high risk counties	8
scenario_decomposition.csv	Three counterfactual scenarios	9
observed_vs_counterfactual.json	Headline framing summary	8, 9
bootstrap_distribution.csv	1,000 bootstrap iterations	10
permutation_importance.csv	Per feature importance shares	11
temporal_validation.json	Yearly trend, Spearman, Jaccard, persistence	12
hrsa_concordance.json	HPSA PC and MH concordance, severity tiers	13
hrsa_concordance_severity.csv	HPSA severity breakdown of 255 high risk	13
federal_cross_reference.json	Three federal anchor sources	14
typology_alignment.csv	Six access typologies	15

typology_concurrence.csv	Multi typology distribution	15
threshold_sensitivity.csv	P65 to P99 sweep	16
state_hei_v1v2_compat.csv	State HEI rankings V1/V2 compatible	18
state_hei_rankings.csv	Alternative state ranking	18

Appendix C. Source File Inventory with SHA 256 Hashes

Label	Path	Size MB	SHA 256 prefix
CMS Geographic Variation 2014 to 2023	Medicare Geographic Variation by NSC/2014-2023 PUF.csv	52.84	658dc3fe c0b08f4b
HRSA HPSA Primary Care	BCD_HPSA_FCT_DET_PC.csv	42.67	fb4e0877 728d6b13
HRSA HPSA Mental Health	BCD_HPSA_FCT_DET_MH.csv	21.40	e65f11efc 725f3c4
AHRF 2024 release	AHRF 2023-2024 CSV/ahrf2024_Feb2025.csv	39.75	99e0e5fd aee611df
AHRF 2023 release	AHRF_CSV_2022-2023/DATA/ ahrf2023.csv	40.10	5b26c17c 0ed99a75
USDA RUCC 2023	Census Demographic Data/Ruralurbancontinuumcodes2023.csv	0.63	ec455ee2 a8bc5fc8

End of preprint manuscript.